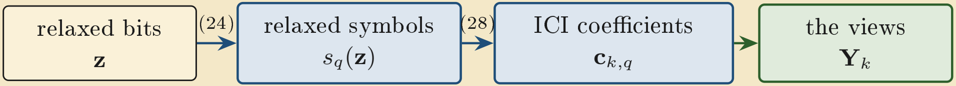
Equations (24)–(30), the receiver forms, and equation (68) of `juna_joint_ieee.tex`

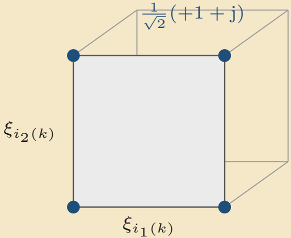
The Joint Demodulation and Decoding Problem

The forward model in ICI coefficients, and one objective over $(\mathbf{C}, \mathbf{z})$



one objective $\mathcal{J}(\mathbf{C}, \mathbf{z})$ scores how well the pair explains the views (30)

The relaxed QPSK symbol

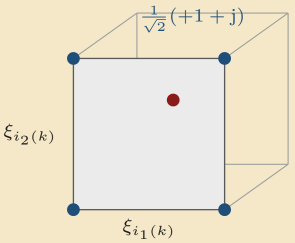


hard bits: $\xi_i = \pm 1$, the four QPSK points

a parity check ties each bit to bits on other carriers; one such bit gives the third axis

Hard bits give the four QPSK points, and relaxed bits move the symbol inside the square those points span. Pilot carriers keep their known symbols. The square is one face of the cube the coded bits span.

The relaxed QPSK symbol



relaxed bits: inside the face

hard bits: $\xi_i = \pm 1$, the four QPSK points

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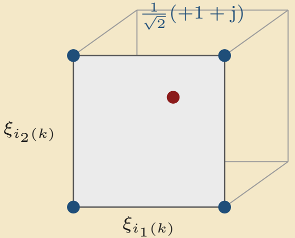

relaxed bits: inside the face

$$s_k(\mathbf{z}) \;=\; \frac{1}{\sqrt{2}}\left[\xi_{i_1(k)}(\mathbf{z}) + \mathrm{j}\,\xi_{i_2(k)}(\mathbf{z})\right]$$

equation (24), where $i_1(k)$ and $i_2(k)$ are the two coded bits of data carrier k and $\xi_i = \tanh(z_i/2)$ is the relaxed bipolar bit of (21)

one relaxed bit gives the real part,
the other the imaginary part

hard bits: $\xi_i = \pm 1$, the four QPSK points

a parity check ties each bit to bits on other carriers; one such bit gives the third axis

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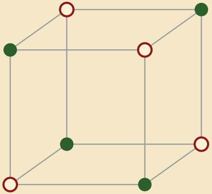
One parity check halves the cube



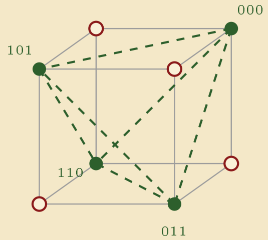
three relaxed bits span the cube $[-1,1]^3$

$$\mathbf{P} = \begin{pmatrix} 1 & 1 & 1 \end{pmatrix}, \qquad \mathbf{P}\mathbf{b}^T = \mathbf{0}$$

codewords 000, 011, 101, 110:
four of the eight corners

Hard bits give the eight cube corners, and one even parity check keeps four. Every cube edge flips one bit and therefore flips parity, so kept corners are never neighbours; they sit on face diagonals. Relaxed bits move inside the cube, and the parity penalty pulls them toward the kept corners.

One parity check halves the cube



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$$\mathcal{R}_{\text{par}}(\mathbf{z}) = (1 - \xi_1\xi_2\xi_3)^2$$

0 at a green corner, 4 at a red one

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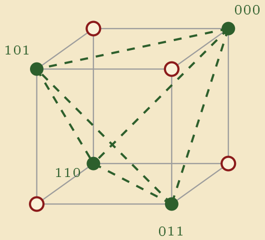

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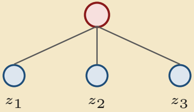
the same check in the Tanner graph



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One coefficient per path

Read $c_{k,q,m} \triangleq \mathbf{e}_k^T \mathbf{F} \mathbf{S}_m \mathbf{G} \mathbf{F}^H \mathbf{e}_q$ from right to left.

$\mathbf{e}_q$ carrier q only → $\mathbf{F}^H$ to time samples → $\mathbf{G}$ the channel acts → $\mathbf{S}_m$ keep segment m → $\mathbf{e}_k^T \mathbf{F}$ FFT, read bin k

One number per triple: transmitted carrier q, received carrier k, view m. The receiver estimates these coefficients; it never evaluates $\mathbf{G}$.

One coefficient per path

Read $c_{k,q,m} \triangleq \mathbf{e}_k^T \mathbf{F} \mathbf{S}_m \mathbf{G} \mathbf{F}^H \mathbf{e}_q$ from right to left.

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$$c_{k,q,m} \triangleq \mathbf{e}_k^T \mathbf{F} \mathbf{S}_m \mathbf{G} \mathbf{F}^H \mathbf{e}_q$$

equation (25): the coupling from transmitted carrier q into view m of carrier k

One number per triple: transmitted carrier q, received carrier k, view m. The receiver estimates these coefficients; it never evaluates $\mathbf{G}$.

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Read $c_{k,q,m} \triangleq \mathbf{e}_k^T \mathbf{F} \mathbf{S}_m \mathbf{G} \mathbf{F}^H \mathbf{e}_q$ from right to left.

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$$c_{k,q,m} \triangleq \mathbf{e}_k^T \mathbf{F} \mathbf{S}_m \mathbf{G} \mathbf{F}^H \mathbf{e}_q$$

equation (25): the coupling from transmitted carrier q into view m of carrier k

$$\nu_{k,m} \triangleq \mathbf{e}_k^T \mathbf{F} \mathbf{S}_m \mathbf{n}$$

equation (26): the same windowed FFT acts on the noise

One number per triple: transmitted carrier q, received carrier k, view m. The receiver estimates these coefficients; it never evaluates $\mathbf{G}$.

Each view is a linear mix

one coefficient per transmitted carrier

$\boxed{c_{k,q_1,m}}$ $\boxed{c_{k,q_2,m}}$ $\boxed{c_{k,q_3,m}}$

$\boxed{X_{q_1}}$ $\boxed{X_{q_2}}$ $\boxed{X_{q_3}}$

the transmitted symbols

Each view is a linear combination of the transmitted symbols plus its windowed noise.

Each view is a linear mix

one coefficient per transmitted carrier

$$\begin{array}{ccc} \boxed{c_{k,q_1,m}} & \boxed{c_{k,q_2,m}} & \boxed{c_{k,q_3,m}} \\ \boxed{X_{q_1}} & \boxed{X_{q_2}} & \boxed{X_{q_3}} \end{array} \Rightarrow \boxed{Y_{k,m}} + \boxed{\nu_{k,m}}$$

the transmitted symbols

view m of carrier k, plus its windowed noise

Each view is a linear combination of the transmitted symbols plus its windowed noise.

Each view is a linear mix

one coefficient per transmitted carrier

$\boxed{c_{k,q_1,m}}$ $\boxed{c_{k,q_2,m}}$ $\boxed{c_{k,q_3,m}}$

$\boxed{X_{q_1}}$ $\boxed{X_{q_2}}$ $\boxed{X_{q_3}}$

$\Rightarrow$ $\boxed{Y_{k,m}} + \boxed{\nu_{k,m}}$

the transmitted symbols view m of carrier k, plus its windowed noise

$$Y_{k,m} = \sum_{q \in \mathcal{K}} c_{k,q,m} X_q + \nu_{k,m}$$

equation (27), where the sum covers the active transmitted carriers q

Each view is a linear combination of the transmitted symbols plus its windowed noise.

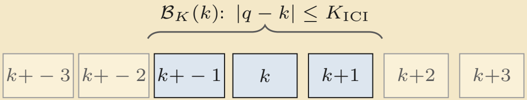
Stack the views, keep the band

$$\boxed{k+-3}\ \boxed{k+-2}\ \overbrace{\boxed{k+-1}\ \boxed{k}\ \boxed{k+1}}^{\mathcal{B}_K(k):\ |q-k|\le K_{\mathrm{ICI}}}\ \boxed{k+2}\ \boxed{k+3}$$

carriers outside the band contribute zero

This is the general forward model of (18) made concrete: the views of carrier k mix only the carriers of its local ICI neighbourhood.

Stack the views, keep the band



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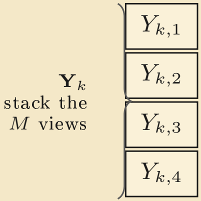

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Stack the views, keep the band

$\mathcal{B}_K(k)$: $|q-k| \le K_{\mathrm{ICI}}$

$k+-3$ $k+-2$ $k+-1$ k $k+1$ $k+2$ $k+3$

carriers outside the band contribute zero

$\mathbf{Y}_k$
stack the
M views

$Y_{k,1}$ $Y_{k,2}$ $Y_{k,3}$ $Y_{k,4}$

$$\mathbf{Y}_k = \sum_{q \in \mathcal{B}_K(k)} \mathbf{c}_{k,q}\, s_q(\mathbf{z}) + \boldsymbol{\nu}_k$$

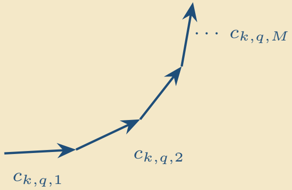
equation (28), where $\mathbf{c}_{k,q} = [c_{k,q,1}, \ldots, c_{k,q,M}]^T$
and the unknown data symbols are
replaced by their relaxed values

linear in the coefficients; the symbols enter through the relaxed bits
$K_{\mathrm{ICI}} = 0$ recovers a per-carrier model

This is the general forward model of (18) made concrete: the views of carrier k mix only the carriers of its local ICI neighbourhood.

The coefficients sum to the channel coefficient

When the windows form a partition of unity, $\sum_m \mathbf{S}_m = \mathbf{I}$.



Component m is one arc of the loop of Fig. 1, and the channel coefficient is the average of the whole loop; the coefficients collect, carrier by carrier, the arcs the views keep.

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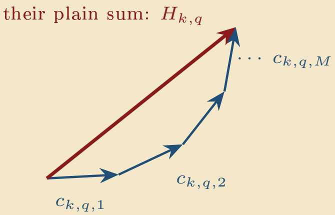

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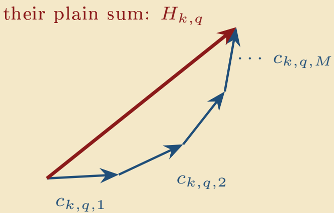

$$\mathbf{1}^T \mathbf{c}_{k,q} \;=\; \sum_{m=1}^{M} c_{k,q,m} \;=\; H_{k,q}$$

equation (29): the M per-view coefficients of a carrier pair sum to its channel coefficient

the same quantity at two resolutions,
exactly as the views sum to the full-FFT bin in (8)

Component m is one arc of the loop of Fig. 1, and the channel coefficient is the average of the whole loop; the coefficients collect, carrier by carrier, the arcs the views keep.

One objective

signal term

$$\lambda_{\mathrm{d}} \sum_{k \in \mathcal{K}} \left\| \mathbf{Y}_k - \sum_{q \in \mathcal{B}_K(k)} \mathbf{c}_{k,q} s_q(\mathbf{z}) \right\|_2^2$$

the Gaussian likelihood of the views under the forward model (28)

Pilot carriers keep their known symbols in $s_q(\mathbf{z})$, so the pilots constrain the coefficients through the signal term. The combiner weights are not variables of this problem.

One objective

signal term

$$\lambda_{\mathrm{d}} \sum_{k\in\mathcal{K}} \left\| \mathbf{Y}_k - \sum_{q\in\mathcal{B}_K(k)} \mathbf{c}_{k,q} s_q(\mathbf{z}) \right\|_2^2$$

coefficient prior

$$\gamma_C \, \|\mathbf{C}\|_F^2$$

the Gaussian likelihood of the views under the forward model (28)

the zero-mean Gaussian prior on $\mathbf{C}$

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parity penalty

$$\lambda_{\mathrm{p}} \, \mathcal{R}_{\mathrm{par}}(\mathbf{z})$$

carried over from (23); vanishes on codewords

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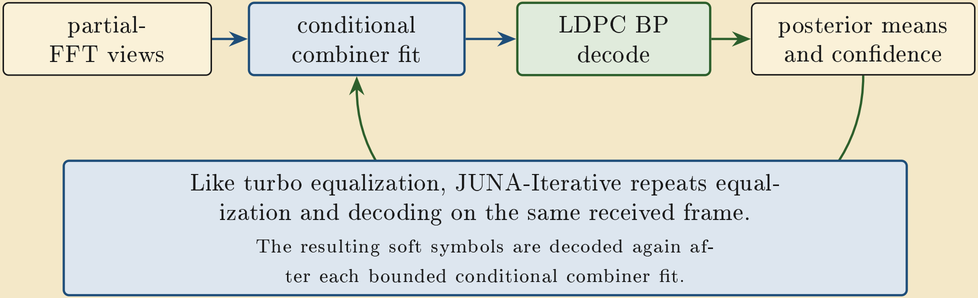
$$\mathcal{J}(\mathbf{C},\mathbf{z}) \quad = \quad \text{signal term} \; + \; \gamma_C \,\|\mathbf{C}\|_F^2 \; + \; \lambda_{\mathrm{p}}\, \mathcal{R}_{\mathrm{par}}(\mathbf{z})$$

equation (30): the general objective (23) with $\boldsymbol{\psi} = \mathbf{C}$;

a function of the coefficients and the relaxed bits alone

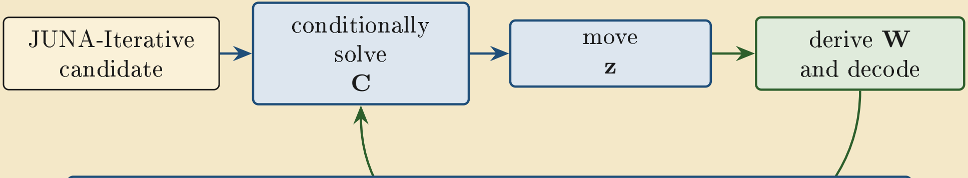
Pilot carriers keep their known symbols in $s_q(\mathbf{z})$, so the pilots constrain the coefficients through the signal term. The combiner weights are not variables of this problem.

JUNA-Iterative



This exchange approximates the joint estimation-and-decoding loop motivated by (30). However, it forms no $\mathbf{C}$*, has no free* $\mathbf{z}$*, and does not use* $\mathcal{J}$ *for step acceptance.*

JUNA-C, z

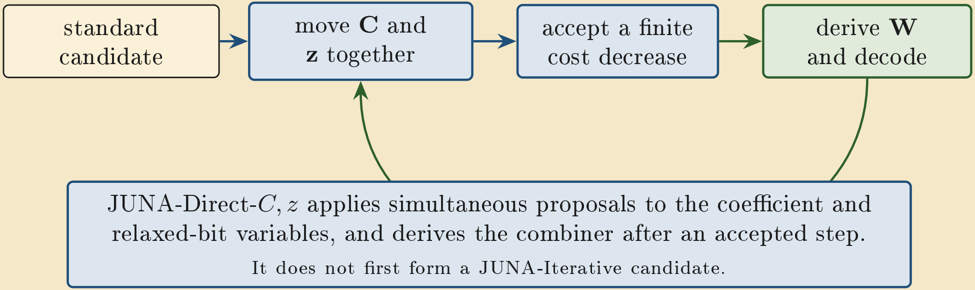


JUNA-C, z conditionally solves the coefficients for the current symbols, derives the read-out weights, and then moves the relaxed bits.

The coefficient and relaxed-bit blocks follow the two variables of the joint objective (30).

This bounded update is motivated by (30). However, the executed rule also contains fixed-reference and decoder terms, so it is not identical to $\mathcal{J}(\mathbf{C}, \mathbf{z})$.

JUNA-Direct-C, z



JUNA-C, z and JUNA-Direct-C, z use the same variables. However, the first conditionally solves the coefficients, whereas the second moves both variables together.

The posterior-variance alternative

the symbol-error model

$$X_q = \mu_q + \varepsilon_q, \text{ with } \mu_q = s_q(\mathbf{z}),$$
$$\mathbb{E}[\varepsilon_q] = 0 \text{ and } \mathbb{E}|\varepsilon_q|^2 = v_q$$

QPSK has unit symbol energy, so $v_q = 1 - |\mu_q|^2$;
on a pilot or another fixed carrier, $v_q = 0$

It is the posterior-variance alternative and is not used for the reported mean-only C,z results; replacing the signal term of (30) by (68) gives the complete expected objective.

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$$\overline{\mathcal{J}}_{\text{sig}} = \lambda_{\text{d}} \sum_{k \in \mathcal{K}} \Big[\Big\| \mathbf{Y}_k - \sum_q \mathbf{c}_{k,q} \mu_q \Big\|_2^2 + \sum_q v_q \left\| \mathbf{c}_{k,q} \right\|_2^2 \Big]$$

equation (68), with both sums over $\mathcal{B}_K(k)$: the expected signal loss under the symbol-error model

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$X_q = \mu_q + \varepsilon_q$, with $\mu_q = s_q(\mathbf{z})$, $\mathbb{E}[\varepsilon_q] = 0$ and $\mathbb{E}|\varepsilon_q|^2 = v_q$

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equation (68), with both sums over $\mathcal{B}_K(k)$: the expected signal loss under the symbol-error model

the first term fits the means; the second retains their uncertainty

the larger v_q, the more strongly the second term bounds that carrier's coefficient energy

It is the posterior-variance alternative and is not used for the reported mean-only C,z results; replacing the signal term of (30) by (68) gives the complete expected objective.

A worked example

$N = 8$ carriers, $M = 4$ views
the channel moves every carrier by a common offset of 0.20 bins

with no offset, $\mathbf{H} = \mathbf{I}$ exactly:
subcarrier orthogonality

Every number in this example is printed by `joint_tutorial_worked_example.py` *beside this file, rounded to four decimals.*

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subcarrier orthogonality

$$|H_{4,4}| = 0.9365$$
$$|H_{4,3}| = 0.2378 \qquad |H_{4,5}| = 0.1618$$

the wanted coefficient shrinks and the neighbouring carriers leak into bin 4

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with no offset, $\mathbf{H} = \mathbf{I}$ exactly:
subcarrier orthogonality

relaxed symbol (24): $\mathbf{z} = (4, -1)$
gives $\xi = (0.9640, -0.4621)$
$s = 0.6817 - 0.3268\mathrm{j}$, so $|s|^2 = 0.5715$
and $v = 1 - |s|^2 = 0.4285$ for (68)

Every number in this example is printed by `joint_tutorial_worked_example.py` *beside this file, rounded to four decimals.*

The example checks (27) and (29)

	$q = 4$ (wanted)	$q = 5$ (neighbour)
$c_{4,q,1}$	$0.2485 + 0.0196\mathrm{j}$	$0.1985 + 0.1011\mathrm{j}$
$c_{4,q,2}$	$0.2303 + 0.0954\mathrm{j}$	$-0.1575 + 0.1575\mathrm{j}$
$c_{4,q,3}$	$0.1895 + 0.1619\mathrm{j}$	$-0.1011 - 0.1985\mathrm{j}$
$c_{4,q,4}$	$0.1302 + 0.2125\mathrm{j}$	$0.2200 - 0.0348\mathrm{j}$
sum	$0.7985 + 0.4893\mathrm{j}$	$0.1598 + 0.0253\mathrm{j}$
$H_{4,q}$	$0.7985 + 0.4893\mathrm{j}$	$0.1598 + 0.0253\mathrm{j}$

The coefficients and the windowed FFT are the same quantity: one is the model, the other the measurement of it.

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(29) checks: the sums equal $H_{4,q}$ to 2.2×10^{-16} over all pairs

the wanted components add nearly in phase; the neighbour components point in different directions and nearly cancel

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(29) checks: the sums equal $H_{4,q}$ to 2.2×10^{-16} over all pairs

the wanted components add nearly in phase; the neighbour components point in different directions and nearly cancel

(27) checks: view 1 of carrier 4 from the windowed FFT is $-0.1396 + 0.3878\mathrm{j}$, and from the coefficient mix $\sum_q c_{4,q,1} X_q$ it is $-0.1396 + 0.3878\mathrm{j}$; they differ by 1.1×10^{-16}

The coefficients and the windowed FFT are the same quantity: one is the model, the other the measurement of it.

The example scores the objective

signal term at the true pair
with noise $\sigma = 0.05$: $\quad 0.0185$
wrong coefficients or wrong bits
make this residual grow

The signal term rewards a pair that reproduces the views, and the parity penalty rewards relaxed bits that harden toward a codeword.

The example scores the objective

signal term at the true pair
with noise $\sigma = 0.05$: 0.0185
wrong coefficients or wrong bits
make this residual grow

one parity check over three bits,
ξ magnitudes (0.9640, 0.4621, 0.9051)

hard bits, even parity	product $+1$	penalty 0
relaxed, correct signs	product $+0.4032$	penalty 0.3561
relaxed, one wrong sign	product -0.4032	penalty 1.9691

The signal term rewards a pair that reproduces the views, and the parity penalty rewards relaxed bits that harden toward a codeword.

The example scores the objective

signal term at the true pair
with noise $\sigma = 0.05$: 0.0185
wrong coefficients or wrong bits
make this residual grow

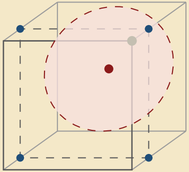
one parity check over three bits,
ξ magnitudes (0.9640, 0.4621, 0.9051)

hard bits, even parity	product $+1$	penalty 0
relaxed, correct signs	product $+0.4032$	penalty 0.3561
relaxed, one wrong sign	product -0.4032	penalty 1.9691

the penalty orders the three cases: satisfied and confident, satisfied but uncertain, violated;
t vanishes exactly on codewords — in the cube, the cases sit at a green corner, inside toward a green corner, and toward a red one

The signal term rewards a pair that reproduces the views, and the parity penalty rewards relaxed bits that harden toward a codeword.

The example takes one gradient step

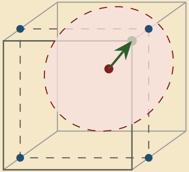


$t = 0$: $z_4 = (0.8000, 0.8000)$
$\xi_4 = (0.3799, 0.3799)$
$v_4 = 0.8556$, $\xi_2 = -0.3799$

carrier 4 sits at
$s_4 = 0.2687 + 0.2687\text{j}$;
$\mathcal{J} = 3.5608$

One plain gradient step on the worked example, printed by the same script. The signal term pulls the symbol toward its corner, the parity penalty pushes the ξ magnitudes toward one, and the carrier's square rides the check bit: the slice slides toward the $\xi_2 = -1$ face while the uncertainty ball shrinks around the triple. The ball is a sphere exactly when the three bits are equally uncertain

The example takes one gradient step



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the gradient at z_4 is
$(-0.2638, -0.2451)$;
with $\eta = 0.8$ the move is
$(+0.2110, +0.1961)$

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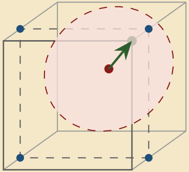

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with $\eta = 0.8$ the move is
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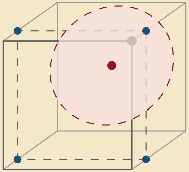
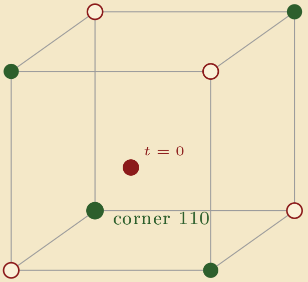
the logits grow, the tanh hardens, the ring shrinks, and $\mathcal{J}$ falls to 2.6663



$t = 1$: $z_4 = (1.0110, 0.9961)$
$\xi_4 = (0.4664, 0.4606)$
$v_4 = 0.7852$, $\xi_2 = -0.4727$

One plain gradient step on the worked example, printed by the same script. The signal term pulls the symbol toward its corner, the parity penalty pushes the ξ magnitudes toward one, and the carrier's square rides the check bit: the slice slides toward the $\xi_2 = -1$ face while the uncertainty ball shrinks around the triple. The ball is a sphere exactly when the three bits are equally uncertain

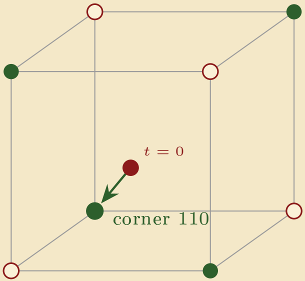
The example steps inside the cube



the check over bits $(2, 3, 8)$;
its codeword corner is 110
$\boldsymbol{\xi} = (-0.3799, -0.3799, 0.3799)$
penalty 0.8933, $\mathcal{J} = 3.5608$

The same gradient step drawn in bit space: the signal term pulls each symbol toward its QPSK corner, and the parity penalty steers the triple toward its codeword corner. The trajectory stays inside the cube because the tanh bounds every ξ.

The example steps inside the cube

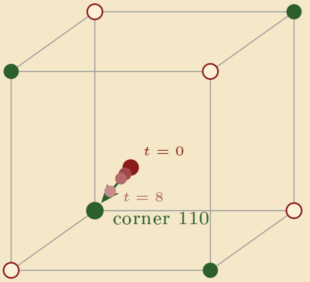


the check over bits $(2, 3, 8)$;
its codeword corner is 110
$\boldsymbol{\xi} = (-0.3799, -0.3799, 0.3799)$
penalty 0.8933, $\mathcal{J} = 3.5608$

the gradient on the triple is
$(0.2839, 0.2765, -0.2638)$;
with $\eta = 0.8$ the move is
$(-0.2271, -0.2212, +0.2110)$
— toward corner 110

The same gradient step drawn in bit space: the signal term pulls each symbol toward its QPSK corner, and the parity penalty steers the triple toward its codeword corner. The trajectory stays inside the cube because the tanh bounds every ξ.

The example steps inside the cube



the check over bits $(2, 3, 8)$;
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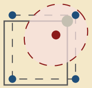
the gradient on the triple is
$(0.2839, 0.2765, -0.2638)$;
with $\eta = 0.8$ the move is
$(-0.2271, -0.2212, +0.2110)$
— toward corner 110

after eight steps
$\boldsymbol{\xi} = (-0.7243, -0.7191, 0.7098)$
penalty 0.3973, $\mathcal{J} = 0.8446$
the penalty falls at every step

The same gradient step drawn in bit space: the signal term pulls each symbol toward its QPSK corner, and the parity penalty steers the triple toward its codeword corner. The trajectory stays inside the cube because the tanh bounds every ξ.

The example shrinks the variance

$t = 0$

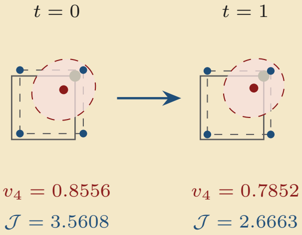


$v_4 = 0.8556$

$\mathcal{J} = 3.5608$

Gradient steps on the worked example, printed by the same script: the coefficients are held at their true values and plain gradient steps with a fixed step size move only $\mathbf{z}$*; the receiver instead solves* $\mathbf{C}$ *conditionally and moves* $\mathbf{z}$ *with Adam. The signal likelihood rises as the objective falls.*

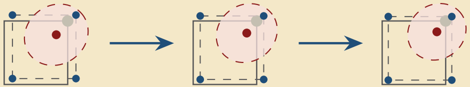
The example shrinks the variance



Gradient steps on the worked example, printed by the same script: the coefficients are held at their true values and plain gradient steps with a fixed step size move only $\mathbf{z}$*; the receiver instead solves* $\mathbf{C}$ *conditionally and moves* $\mathbf{z}$ *with Adam. The signal likelihood rises as the objective falls.*

The example shrinks the variance

$t = 0$ | $t = 1$ | $t = 2$



$v_4 = 0.8556$, $\mathcal{J} = 3.5608$

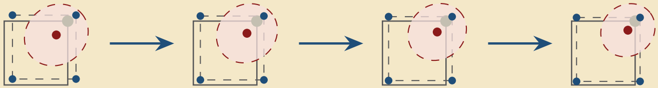
$v_4 = 0.7852$, $\mathcal{J} = 2.6663$

$v_4 = 0.7241$, $\mathcal{J} = 2.0942$

Gradient steps on the worked example, printed by the same script: the coefficients are held at their true values and plain gradient steps with a fixed step size move only $\mathbf{z}$*; the receiver instead solves* $\mathbf{C}$ *conditionally and moves* $\mathbf{z}$ *with Adam. The signal likelihood rises as the objective falls.*

The example shrinks the variance

$t = 0$	$t = 1$	$t = 2$	$t = 4$

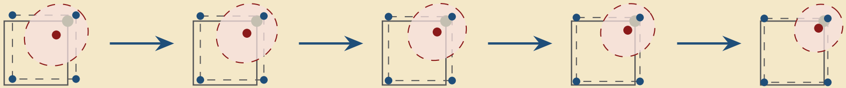


$v_4 = 0.8556$	$v_4 = 0.7852$	$v_4 = 0.7241$	$v_4 = 0.6286$
$\mathcal{J} = 3.5608$	$\mathcal{J} = 2.6663$	$\mathcal{J} = 2.0942$	$\mathcal{J} = 1.4296$

Gradient steps on the worked example, printed by the same script: the coefficients are held at their true values and plain gradient steps with a fixed step size move only $\mathbf{z}$*; the receiver instead solves* $\mathbf{C}$ *conditionally and moves* $\mathbf{z}$ *with Adam. The signal likelihood rises as the objective falls.*

The example shrinks the variance

$t = 0$	$t = 1$	$t = 2$	$t = 4$	$t = 8$

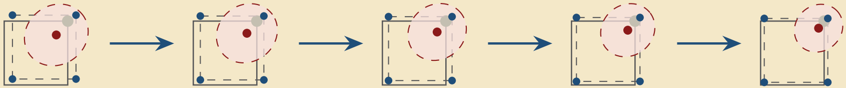


$v_4 = 0.8556$	$v_4 = 0.7852$	$v_4 = 0.7241$	$v_4 = 0.6286$	$v_4 = 0.5067$
$\mathcal{J} = 3.5608$	$\mathcal{J} = 2.6663$	$\mathcal{J} = 2.0942$	$\mathcal{J} = 1.4296$	$\mathcal{J} = 0.8446$

Gradient steps on the worked example, printed by the same script: the coefficients are held at their true values and plain gradient steps with a fixed step size move only $\mathbf{z}$*; the receiver instead solves* $\mathbf{C}$ *conditionally and moves* $\mathbf{z}$ *with Adam. The signal likelihood rises as the objective falls.*

The example shrinks the variance

$t = 0$ | $t = 1$ | $t = 2$ | $t = 4$ | $t = 8$



$v_4 = 0.8556$	$v_4 = 0.7852$	$v_4 = 0.7241$	$v_4 = 0.6286$	$v_4 = 0.5067$
$\mathcal{J} = 3.5608$	$\mathcal{J} = 2.6663$	$\mathcal{J} = 2.0942$	$\mathcal{J} = 1.4296$	$\mathcal{J} = 0.8446$

every one of the eight steps lowers $\mathcal{J}$, v_4, and the mean v, which falls from 0.8556 to 0.4816

descent hardens the relaxed bits: $|\mu_q|$ grows, so $v_q = 1 - |\mu_q|^2$ falls

Gradient steps on the worked example, printed by the same script: the coefficients are held at their true values and plain gradient steps with a fixed step size move only $\mathbf{z}$*; the receiver instead solves* $\mathbf{C}$ *conditionally and moves* $\mathbf{z}$ *with Adam. The signal likelihood rises as the objective falls.*

The joint demodulation and decoding problem

relaxed bits form the symbols (24); the ICI coefficients carry them into the views (25)–(28)

the per-view coefficients sum to the channel coefficient (29)

one objective scores the pair $(\mathbf{C}, \mathbf{z})$ (30)

JUNA-Iterative exchanges a conditional combiner fit and decoding; JUNA-C, z and JUNA-Direct-C, z move the joint variables

the posterior-variance alternative is equation (68)

the pilots constrain the coefficients; the parity checks constrain the bits; both act in the same objective